② $V_z = 6V$

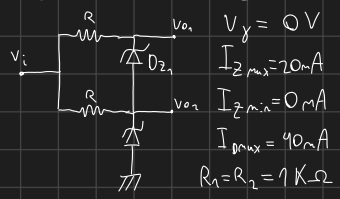
$V_\gamma = 0V$

$I_{Z\,max} = 20mA$

$I_{Z\,min} = 0mA$

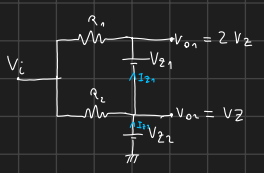
$I_{D\,max} = 40mA$

$R_1 = R_2 = 1K\Omega$



a) Zona funcionamiento?

$V_{O_1} = 12V \quad V_{O_2} = 6V$



$I_{Z_1} = I_{R_1} = \frac{2V_z - V_i}{R_1} \;//\; -I_{Z\,max} \le I_{Z_1} \le -I_{Z\,min} \;//\; -20\cdot10^{-3} \le \frac{2V_z - V_i}{R_1} \le 0$

$I_{Z_2} + I_{R_2} = I_{R_1} \;//\; I_{Z_2} = I_{R_1} - I_{R_2} = \frac{2V_z - V_i}{R_1} + \frac{V_z - V_i}{R_2} = \frac{3V_z - 2V_i}{R_1} \;//\; -I_{Z\,max} \le I_{Z_2} \le -I_{Z\,min} \;//$

$-20\cdot10^{-3} \le \frac{3V_z - 2V_i}{R_1} \le 0$

$I_{R_2} = \frac{V_i - V_z}{R_2}$

$I_{Z_1} \to -20 \le 12 - V_i \le 0 \;//\; -32 \le -V_i \le -12 \;//\; \boxed{32 \ge V_i \ge 12}$

$I_{Z_2} \to -20$

b) ¿Zonas funcionamiento?

$V_{O_1} = V_i \;//\; V_{O_2} = 6V$

